METRIC

DOD-B-70331(AR)
21 April 1983

MILITARY SPECIFICATION

BOLT, SELF-LOCKING, HEXAGON HEAD, STEEL, METRIC,
GENERAL SPECIFICATION FOR

This specification is approved for use by US Army Armament Research and Development Command Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of hexagon head, self-locking bolts, property class 10.9 steel, with metric dimensions.

1.2 Classification. Bolts shall be of the following types as specified on the applicable specification sheets (see 6.2).

Type B - Standard Hex Head
Type BH - Larger Hex Head

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ, 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5306

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DOD-B-70331/1 - Bolt, Self-Locking, Hexagon Head, Steel, Type B, Metric
DOD-B-70331/2 - Bolt, Self-Locking, Hexagon Head, Steel, Type BH, Metric
DOD-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base, (For Ferrous Metals)

STANDARDS

FEDERAL

FED-STD-H28/21 - Screw Thread Standards for Federal Services, Section 21, Metric Screw Threads

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-1312 - Fasteners, Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

SAE J1199 (10.9) - Mechanical and Material Requirements for Metric Externally Threaded Steel Fasteners

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 19096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. The bolts shall conform to the material composition of SAE J1199 (10.9).

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3.3 Design. The bolts shall conform to the design, dimensions, and tolerances specified on specification sheets.

3.4 Finish. The protective finish of the bolts when specified in the applicable specification sheet shall be as follows (see 4.3.4).

3.4.1 Cadmium plate. Bolts shall be cadmium plated in accordance with QQ-P-416, type II, class 2. Plating thickness shall be 7.6 μm (micrometers) thick.

3.4.2 Phosphate coating. Bolts shall be phosphate coated in accordance with DOD-P-16232, type Z, class 2. Coating thickness shall be 5 to 15 μm (micrometers) thick.

3.4.3 Hydrogen embrittlement. Cadmium plated bolts shall be subjected to an embrittlement relief treatment in accordance with QQ-P-416 as soon as possible after plating to minimize the resulting embrittlement.

3.5 Mechanical properties. The mechanical properties of the bolts shall meet the requirements as specified in table I (see 4.3.2 and 4.2.4).

TABLE I. Mechanical properties.

Nom thread dia and thread pitch	Stress area mm^2	Proof load kN	Tensile strength min kN
M6x1	20.1	16.7	20.9
M8x1.25	36.6	30.4	38.1
M10x1.5	58.0	48.1	60.3
M12x1.75	84.3	70.0	87.7
M14x2	115.0	95.4	120.0
M16x2	157.0	130.0	163.0
M20x2.5	245.0	203.0	255.0
M24x3	353.0	293.0	367.0
M30x3.5	561.0	466.0	583.0

3.5.1 Hardness. Bolts shall have a hardness of 33-39HRC.

3.6 Threads. Threads shall be standard course pitch "M" profile, tolerance class 6g after plating, as specified on the applicable specification sheets and in accordance with FED-STD-H28/21. Threads shall be formed by rolling process.

3.7 Head.

3.7.1 Head taper. The taper of the head (angle between one side or sides of the head and the axis of the fastener) shall not exceed 2 degrees.

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3.7.2 Bearing surface. The bearing surface of the head shall be at right angles to the axis of the body within 2 degrees for bolt sizes up to and including 24 mm and within 1 degree for bolt sizes larger than 24 mm. The bearing surface of the head shall be concentric with the axis of the body within 3 percent of the maximum width across the flats. The diameter of bearing surface shall be equal to 95 percent of maximum width across flats within plus or minus 5 percent.

3.8 Workmanship. Finished bolts shall be uniform in quality and appearance. Surfaces shall be free of foreign matter. Bolt surfaces in the unplated condition shall be free of cracks, bursts, folds, tool marks, seams, voids on the bearing surface, nicks and gouges.

3.9 Marking. Bolts shall be marked with military part number and manufacturer's identification in accordance with MIL-STD-130.

3.10 Self-locking feature. The self-locking feature of the bolts, shall be fully effective when tightened to the torque values specified in table II. Where material softer than the bolt material is fastened, a plain washer of material as hard as the bolt shall be used under the bolt head. Bolts shall successfully meet vibration testing without loosening of bolt and nut assembly nor show evidence of cracks or breaks when tested in accordance with 4.3.5.

TABLE II. Torque values. 1/

Nominal thread diameter	N.m	
	Max	Min
M6X1.00	15	12
M8X1.25	37	28
M10X1.50	74	56
M12X1.75	129	98
M14X2	204	156
M16X2	318	243
M20X2.5	621	475
M24X3	1075	823
M30X3.5	2138	1635

1/ Torque values based on 0.18 coefficient of friction and 85% of proof load (max), 65% of proof load (min).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in table III.

TABLE III. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Group A Examination	1/	4.3.1
Group B Mechanical properties Hardness Self-locking feature	 3.5 3.5.1 3.10	 4.3.2 4.3.2 4.3.5

1/ All requirements not covered by tests.

4.2.1 Inspection lot. An inspection lot shall consist of all bolts covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.2.3 Sampling for group A inspection. Sampling shall be in accordance with MIL-STD-105 inspection level II with an acceptable quality level (AQL) of 1.5 percent defective for major defects. For minor defects, inspection level I with an AQL of 4.0 percent defective shall apply.

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4.2.3.1 Classification of defects. Defects shall be classified as specified in table IV.

TABLE IV. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major		
101	Threads, not as specified (see 3.6)	Gage
102	Undercut depth of head (see 3.1)	SIE 1/
103	Width across flats, not as specified (see 3.1)	SIE
104	Slot and recess radius of head (see 3.1)	SIE
105	Bearing surface of head (see 3.7.2)	SIE
106	Self-locking feature (see 3.10)	SIE
Minor		
201	Overall length of bolt, not as specified (see 3.1)	SIE
202	Thread length (see 3.1)	SIE
203	Other dimensions, not as specified (see 3.1)	SIE
204	Protective finish missing (see 3.4)	Visual
205	Fillet under head, not as specified (see 3.1)	SIE
206	Head taper, not as specified (see 3.7.1)	SIE
207	Bearing surface-bolt axis perpendicularity, not as specified (see 3.7.2)	SIE
208	Workmanship, not as specified (see 3.8)	Visual
209	Packaging, not as specified (see 5.1)	Visual

1/ Standard inspection equipment

4.2.4 Sampling for group B inspection. Sampling shall be in accordance with MIL-STD-105 inspection level S-2 with an AQL of 1.5 percent defective.

4.2.4.1 Disposition of sample units. Bolts which have been subjected to group B inspection shall not be delivered on the contract.

4.2.5 Packaging inspection. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with PPP-H-1581.

4.3 Methods of inspection.

4.3.1 Examination. Sample bolts shall be thoroughly examined to determine conformance with this specification and applicable specification sheets with respect to all the requirements not covered by tests.

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4.3.2 Mechanical properties inspection. Sample bolts shall be tested for conformance to requirements of 3.5 and 3.5.1. Tensile strength shall be determined in accordance with MIL-STD-1312, test method 8. Surface hardness shall be determined in accordance with MIL-STD-1312, test method 6.

4.3.3 Materials inspection. Material inspection shall consist of certification supporting verifying data that materials used in fabricating bolts are in accordance with 3.2, 3.4, and 3.5.

4.3.4 Protective finish inspection. Sample bolts shall be inspected for minimum thickness and continuity of the plating in accordance with QQ-P-416 or DOD-P-16232, as applicable (see 3.4).

4.3.5 Self-locking feature test method. The self-locking feature, specified in 3.10, shall be subjected to vibration test of MIL-STD-1312,Test 7. Sample bolts 16 mm diameter and smaller shall be installed in test fixture with flat washers and nuts and tightened to the torque values specified in table II. Loosening of bolt and nut assembly, cracks or breaks during vibration life of less than fifteen (15) minutes shall be cause for rejection. Vibration test on bolts over 16 mm diameter are waived provided the 16 mm and smaller of the same type and design have satisfactorily passed the test.

5. PACKAGING

5.1 Packaging requirements. The bolts shall be prepared for shipment as specified in PPP-H-1581.

5.2 Preservation. The level of protection required (levels A, B, or C) shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The bolt covered by this specification is intended for use when a self-locking fastener is needed or where the reduction of fatigue failure is desired. Typical applications include uses as cylinder head bolts, flywheel housing bolts, connecting rod bolts, and as bearing capscrews. The bolt may be used in tapped holes or with nuts.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1).
c. Level (degree) of protection, in accordance with PPP-H-1581 (see 5.2).

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6.3 Definitions.

6.3.1 Description. The bolt covered by this specification is a one piece, free-spinning type of self-locking bolt. Locking action is provided by a flexible diaphragm which is formed between the slotted segments of the upper part of the bolt head and the circular recess adjacent to the shank in the lower part of the head. This diaphragm acts as a controlled spring element when the head is wrenched against a rigid seat. When used as a bearing cap screw, the material against which the head of the bolt seats should be hard enough to form a rigid seat.

6.4 Patent notice. This bolt is covered by U.S. Patent Number 3342236, which expires 19 September 1984. The Government does not have a royalty free license under this patent. This bolt is known as a "place" bolt under this patent.

Custodians:
Army - AR

Review activities:
Army - AR, MI
DLA - IS
NSA - NS

User activities:
Army - AT, GL, ME
Navy - MC, OS, SH

Agent:
DLA - IS

Preparing activity:
Army - AR

(Project 5306-0748)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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